Operating system

An **Operating System** acts as a communication bridge (interface) between the user and computer hardware. The purpose of an operating system is to provide a platform on which a user can execute programs in a convenient and efficient manner.

An operating system is a piece of software that manages the allocation of computer hardware. The coordination of the hardware must be appropriate to ensure the correct working of the computer system and to prevent user programs from interfering with the proper working of the system.

Functions of an operating System:

- **Security –**
  The operating system uses password protection to protect user data and similar other techniques. it also prevents unauthorized access to programs and user data.
- **Control over system performance –**
  Monitors overall system health to help improve performance. records the response time between service requests and system response to have a complete view of the system health. This can help improve performance by providing important information needed to troubleshoot problems.
- **Job accounting –**
  Operating system Keeps track of time and resources used by various tasks and users, this information can be used to track resource usage for a particular user or group of user.
- **Error detecting aids –**
  Operating system constantly monitors the system to detect errors and avoid the malfunctioning of computer system.
- **Coordination between other software and users –**
  Operating systems also coordinate and assign interpreters, compilers, assemblers and other software to the various users of the computer systems.

Memory Management –
The operating system manages the Primary Memory or Main Memory. Main memory is made up of a large array of bytes or words where each byte or word is assigned a certain address. Main memory is a fast storage and it can be accessed directly by the CPU. For a program to be executed, it should be first loaded in the main memory. An Operating System performs the following activities for memory management: It keeps tracks of primary memory, i.e., which bytes of memory are used by which user program.

Processor Management –
In a multi programming environment, the OS decides the order in which processes have access to the processor, and how much processing time each process has. This function of OS is called process scheduling. An Operating System performs the following activities for processor management. Keeps tracks of the status of processes. The program which perform this task is known as traffic controller. Allocates the CPU that is processor to a process. De-allocates processor when a process is no more required.

Device Management –
An OS manages device communication via their respective drivers. It performs the following activities for device management. Keeps tracks of all devices connected to system. designates a program responsible for every device known as the Input/output controller. Decides which process gets access to a certain device and for how long. Allocates devices in an effective and efficient way. De allocates devices when they are no longer required.

File Management –

A file system is organized into directories for efficient or easy navigation and usage. These directories may contain other directories and other files. An Operating System carries out the following file management activities. It keeps track of where information is stored, user access settings and status of every file and more… These facilities are collectively known as the file system.

Operating System design Approaches

Layered Approach

A system can have different designs and modules. One of them is the layered approach, in which the operating system is broken into a number of layers, the bottom layer (layer 0) being hardware and the highest (layer N) being the user interface.

An operating system layers have the abstraction of data and functions within a layer. In a typical Layer system, say layer M, consists of data structures and a set of routines that can be invoked by higher-level layers and in turn Layer M, can call operations on lower-level layers.

In this approach, construction is simple in understanding and debugging. If an error is found during the debugging of a particular layer, the error must be on that layer, because the layers below it are already debugged. Thus, the design and implementation of the systems are simplified.

Virtual Machine (VM) Approach

A Virtual Machine can share the same hardware on several different execution environments concurrently. It is one of the main advantages of using VM. Another important advantage is that the host system is protected from the virtual machines, just like virtual machines are protected from each other. A virus inside a guest operating system might damage that operating system but is unlikely to affect the host or the other guests. This is because each virtual machine is completely isolated from all other virtual machines.

Kernel-Based Approach

A kernel is a central component of an operating system. It acts as an interface between the user applications and the hardware. It manages the communication between the software (user level applications) and the hardware (CPU, disk memory, etc). The kernel provides functions like Process management, Device management, Memory management, Interrupt handling, I/O communication, File system, etc.

Types Of Advance Operating System

Network Operating System (NOS)

A Network operating system provides an environment in which users, who are aware of the multiplicity of machines, can access remote resources by either logging in to the appropriate remote machine or transferring data from the remote machine to their own machines.

1.1 Remote Login

An important function of a network operating system is to allow users to log in remotely. The Internet provides the telnet facility for this purpose.

1.2 Peripheral Sharing Service

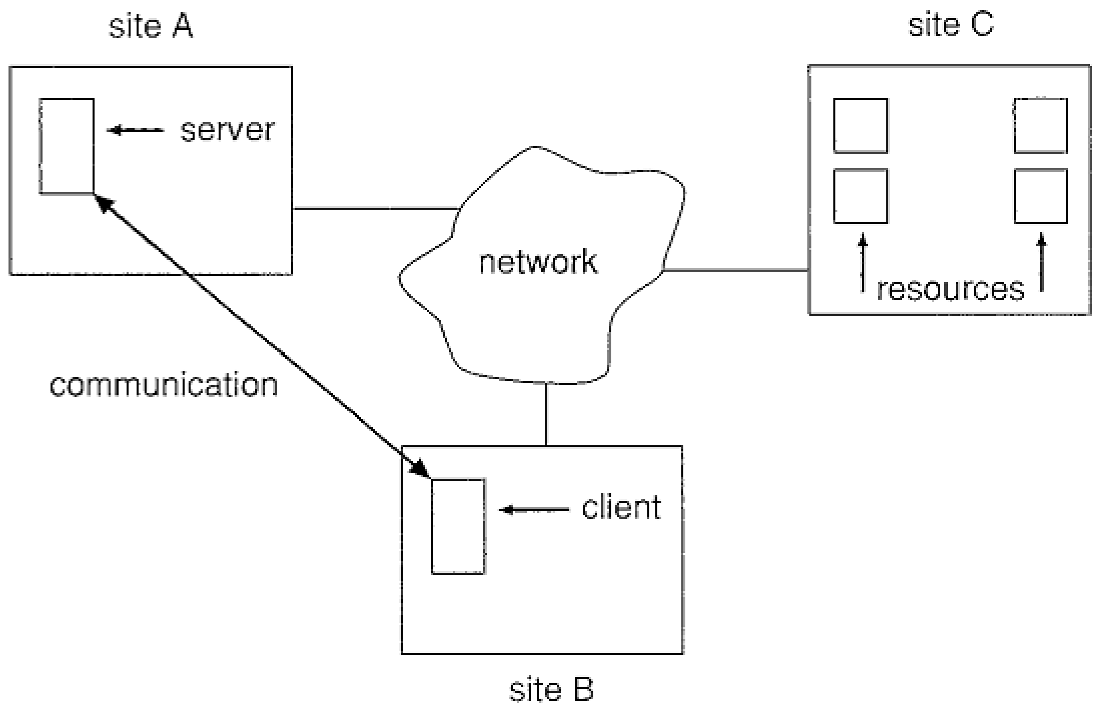
Peripherals connected to one computer are often shared by other computers, by the use of peripheral sharing services. These services go by many names, such as remote device access, printer sharing, shared disks and so on. A computer having a peripheral device makes it available by exporting it. Other computers can connect to the exported peripheral. After a connection is made, to a user on the machine connected to a shared peripheral, that peripheral appears to be local (that is, connected to the users machine).

1.3 Remote File Transfer Service

The most common service that a network operating system provides is file service. File services allow user of a set of computers to access files and other persistent storage object from any computer connected to the network. The files are stored in one or more machines called the file server(s). The machines that use these files, often called workstations have transparent access to these files.

2 Distributed Operating System(DOS)

In a distributed operating system, users access remote resources in the same way they access local resources.Data and process migration from one site to another is under the control of the distributed operating system. The processors in a distributed system may vary in size and function. They may include small microprocessors,workstations,minicomputers,and large general-purpose cornputer systems. These processors are referred to by a number of names, such as sites, nodes, computers, machines, and hosts, depending on the context in which they are mentioned. We mainly use site to indicate the location of a machine and host to refer to a specific system at a site. Generally, one host at one site, the server, has a resource that another host at another site, the client (or user), would like to use.


site A
server
site C
network
resources
communication
client
site B

There are four major reasons for building distributed systems: resource sharing, computation speedup, reliability, and communication. In this section, we briefly discuss each of them.

Resource Sharing

If a number of different sites (with different capabilities) are connected to one another, then a user at one site may be able to use the resources available at another. For example, a user at site A may be using a laser printer located at site B. Meanwhile, a user at B may access a file that resides at A.

Computation Speedup

If a particular computation can be partitioned into sub computations that can run concurrently, then a distributed system allows us to distribute the sub computations among the various sites; the sub computations can be run concurrently and thus provide computation speedup.

Reliability

If one site fails in a distributed system, the remaining sites can continue operating, giving the system better reliability. If the system is composed of multiple large autonomous installations (that is, general-purpose computers), the failure of one of them should not affect the rest.

Real Time Operating System(RTOS)

A real-time system is a computer system that requires not only that the computing results be "correct" but also that the results be produced within a specified deadline period. Results produced after the deadline has passed-even if correct-may be of no real value. Real-time systems executing on traditional computer hardware are used in a wide range of applications. In addition, many real-time systems are embedded in "specialized devices," such as ordinary home appliances (for example, microwave ovens and dishwashers), consumer digital devices (for example, cameras and MP3 players), and communication devices (for example, cellular telephones and Blackberry handheld devices).

Real-time computing is of two types: hard and soft. A hard real-time system has the most stringent requirements, guaranteeing that critical real-time tasks be completed within their deadlines. Safety-critical systems are typically hard real-time systems. A soft real-time system is less restrictive, simply providing that a critical real-time task will receive priority over other tasks and that it will retain that priority until it completes.

The following characteristics are typical of many real-time systems:

Single purpose

Small size

Inexpensively mass-produced

Specific timing requirements

Mobile OS(MOS)

The operating system operates on a mobile device and is called mobile operating system.

In total it shows up 16 different mobile operation systems, whereof twelve are actual on the market or will be released in an expected time. An exception is the operation system MeeGoo of the companies Nokia and IBM, which future is unknown after Nokia announced that they will use the operating system Windows Phone for their new devices.

Mobile OS: Android

Android is an open source operating system for mobile devices developed by Google and the Open Handset Alliance. With 22,7% it is the second most used operating system for mobile devices worldwide behind Symbian.

The system architecture consists of :

A modified Linux Kernel

open source Libraries coded in C and C++

The Android Runtime, which considers core libraries that disposals the most core functions of Java. As virtual machine it uses Dalvin, which enables to execute Java applications.

An Application Framework , which disposals services and libraries coded in Java for the application development

Cloud Operating System(COS)

A Cloud is a logical entity composed of managed computing resources deployed in private facilities and interconnected over a public network, such as the Internet. Cloud machines (also called nodes) are comprised of inexpensive, off-the-shelf consumer-grade hardware. Clouds are comprised of a large number of clusters (i.e. sets of nodes contained in a same facility) whose size may range from a few machines to entire datacenters.

The Cloud OS aims to provide a familiar interface for developing and deploying massively scalable distributed applications on behalf of a large number of users, exploiting the seemingly infinite CPU, storage, and bandwidth provided by the Cloud infrastructure. The features of the Cloud OS aim to be an extension to those of modern operating systems, such as UNIX and its successors: in addition to simple programming abstractions and strong isolation techniques between users and applications, we emphasize the need to provide a much stronger level of integration with network resources.

Multiprocessor Operating System

A computer system in which two or more CPUs share full access to a common RAM called multiprocessor OS.